Advanced Mathematical Techniques Worksheet 4
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The purpose of this worksheet is to give you some practice working with expansions of the Gamma function and using these expansions to determine the value of various integrals. You may use Mathematica or another computer algebra system to check your work, but please do all of your work by hand.

1. This problem concerns the evaluation of integrals of the form $\int_0^\infty e^{-ax} \ln^n x \, dx$ for natural numbers n and positive real numbers a.

   (a) Use the integral representation of the Gamma function to write the value of the integral $\int_0^\infty x^\varepsilon e^{-x} dx$ in terms of Γ.

   (b) Write the Maclaurin expansion for x^ε, thought of as a function of ε rather than x. Your expansion should contain powers of ε.

   (c) Substitute your result from part (b) into the integral given in part (a) and exchange the order of summation and integration to obtain a series of the form $\sum_{k=0}^{\infty} c_k \varepsilon^k$, where the coefficients c_k contain integrals involving x. It is not necessary to justify the exchange of summation and integration for this problem.

   (d) Use the Maclaurin expansion for $\ln\Gamma(1+x)$ derived in class to determine the first four nonzero terms in the Maclaurin expansion for $\Gamma(1+\varepsilon)$, thought of as a function of ε. Your coefficients may be left in terms of the Zeta function at integer arguments and/or the Euler-Mascheroni constant γ.

   (e) Combine your results from parts (a), (c), and (d) to determine the values of the integrals

   (i) $\int_0^\infty e^{-x} \ln x \, dx$. (ii) $\int_0^\infty e^{-x} \ln^2 x \, dx$. (iii) $\int_0^\infty e^{-x} \ln^3 x \, dx$.

   (f) Re-do part (a) for the integral $\int_0^\infty x^\varepsilon e^{-ax} dx$, assuming a is a positive real number. Then, use your result from part (d) to find the first four nonzero terms in a Maclaurin series for this integral in powers of ε. Hint: this can be done very easily with a few observations. Note that $7^\varepsilon = e^{\varepsilon \ln 7}$ so $7^\varepsilon e^{4\varepsilon} = e^{(4+\ln 7)\varepsilon}$.

   (g) Argue that there is no substantive difference between performing the manipulation in part (c) for $\int_0^\infty x^\varepsilon e^{-x} dx$ and performing that manipulation for $\int_0^\infty x^\varepsilon e^{-ax} dx$. Then, use your result from part (f) in conjunction with this fact to evaluate $\int_0^\infty e^{-5x} \ln^2 x \, dx$.

2. This problem concerns the evaluation of the integral $\int_0^\infty xe^{-3x}\ln^2 x\,dx$.

   (a) Evaluate the integral $\int_0^\infty x^{1+\varepsilon}e^{-3x}\,dx$ in terms of the Gamma function.

   (b) Use the recurrence relation for the Gamma function to re-write your result from part (a) in a form that allows you to make use of the expansion found in part (d) of problem 1.

   (c) Write out the first three nonzero terms of the Maclaurin expansion for the integral you found in part (a).

   (d) Use the manipulations you performed in parts (b) and (c) of problem 1 to write a Maclaurin expansion of the form $\sum_{k=0}^{\infty} c_k\varepsilon^k$ for the integral in part (a). Your coefficients will contain integrals involving *x*. Note that you do *not* want powers of $(1+\varepsilon)$. You want powers of ε.

   (e) Use your results from parts (c) and (d) along with those from problem 1 to determine the value of the integral $\int_0^\infty e^{-3x}x\ln^2 x\,dx$. Your result should be written in terms of the irrational numbers γ, π, and $\ln 3$ only, along with rational numbers.

3. Use the integral expression for the Beta function derived in class to determine the value of the following integrals. You may include $\pi, \sqrt{3}, \Gamma(1/3)$, and $\Gamma(1/4)$ in your expressions, but no other irrational numbers (note that this means $\Gamma^2(1/3)$ is allowable, but $\Gamma(2/3)$ and γ are not).

   (a) $\int_0^1 \sqrt{\frac{t}{1-t}}dt$. (b) $\int_0^1 t^4\sqrt{1-t^2}\,dt$. (c) $\int_0^1 \sqrt[3]{t^2(1-t)}\,dt$.